

MD MIRAJ AREFIN

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RESEARCH INTERESTS

Medical Robotics Robot Learning Robot Dynamics and Control Motion Planning

EDUCATION

Middle East Technical University (METU), Turkey Sep 2025 - Aug 2027
MS in Robotics **CGPA: 2.90/4(15/21 credits)**
Advisor : Dr. Mehmet Efe Tiryaki
Thesis: Autonomous Robot-assisted echocardiography system design

Khulna University of Engineering & Technology (KUET), Bangladesh Jan 2018 - June 2024
B.Sc. in Mechanical Engineering **CGPA: 2.62/4**

RESEARCH EXPERIENCE

Robotic Ultrasound Imaging System Oct 2025 - Present
Graduate Research Assistant, Medical Robotic Systems Lab, METU
Supervisor: Dr. Mehmet Efe Tiryaki

- Developed B-mode ultrasound contact detection and localization using intensity profiles and curved-probe geometry, independently of force measurements.
- Formulated image-guided, contact-aware impedance control with an updated interaction frame and contact-conditioned force/torque taring.
- Validated on a Franka Research 3 robot arm and agar phantom, achieving contact-localization RMSE of 1.46 ± 0.14 mm against a geometric reference and reducing peak compressive force by **36.3%** versus a fixed-frame baseline.
- Demonstrated 50 mm scans at varied probe orientations, with along-axis RMSE of 0.93–1.60 mm and surface-normal RMSE below 0.50 mm.
- Developed predictive admittance control with online tissue-stiffness estimation and target-depth prediction on a UR5e robot arm, achieving **0.44 N force RMSE** after stabilization at a 10 N target and executing continuous 3D square-spiral scans.

Fixed Wing VTOL for Emergency Medical Vaccine Supply Sep 2023 – Mar 2024
Undergraduate Research Assistant, KUET
Supervisor: Dr. Md. Ashraful Islam

- Designed and constructed a hybrid fixed-wing VTOL aircraft (8 kg take-off weight) for emergency medical transport applications.
- Developed a modular airframe achieving successful VTOL flight testing with approximately 5 minutes endurance.

PUBLICATIONS

Submitted Manuscript

- [S1] **Miraj Arefin** and M. E. Tiryaki. “Contact-Aware Impedance Controller for Robot-Assisted Ultrasound Imaging.” Submitted to **IEEE ICRA 2027**. [[Research overview](#)]

Conference Publications

- [C4] **Miraj Arefin** and M. E. Tiryaki. “Tissue Stiffness Predictive Admittance Control for Robot-Assistive Ultrasound Imaging.” *7th Turkish Robotics Conference*, 2026. Poster. [[Poster](#)]
- [C3] A. H. Arif, M. M. Abedin, S. Das, S. E. Faruquei, A. Rahaman, and **Miraj Arefin**. “Automated Kidney Tumor Detection and Segmentation Using YOLO11 and Grad-CAM.” *International Informatics and Software Engineering Conference (IISEC)*, 2026. [[DOI](#)]
- [C2] M. T. Rana, **Miraj Arefin**, N. Sharmin, H. A. Begum, M. F. Raihan, and A. Rahman. “Dynamic Modeling and Propulsion System Analysis with LQR-Based Control System for Enhanced Performance of a Lighter-Than-Air Aerial Drone.” *DELTA's* 2024. [[DOI](#)]
- [C1] N. S. Mooaz, **Miraj Arefin**, S. R. Dhrubo, and A. Ahmed. “Transforming UAV Design: Coandă-Based Lift Generation for Enhanced Aerial Performance.” *International Conference on Electrical Information and Communication Technology (EICT)*, 2023. [[DOI](#)]

Journal publications

[J3] M. F. Rabbi, R. Sarker, J. Becker, A. R. Badsha, **Miraj Arefin**, I. Zahan, M. Arifuzzaman, M. A. Al Bari, and M. G. Kibria. “Investigating the Shear and Thermal Behavior of Additively Manufactured Wood-PLA Biocomposites Including a Life Cycle Assessment.” *Journal of Natural Fibers*, vol. 23, no. 1, article 2640270 (2026). [DOI]

[J2] M. M. M. Haque, S. R. Dhrubo, A. Z. Pranto, A. Ahmed, **Miraj Arefin**, M. Arifuzzaman, and M. S. Islam. “Impact of Process Parameters and Material Selection on the Mechanical Performance of FDM 3D-Printed Components.” *Hybrid Advances* (2025). [DOI]

[J1] **Miraj Arefin**, D. Mondal, and M. A. Islam. “Thermodynamic Analysis of Cascade Refrigeration System using Low GWP Refrigerants for Low-Temperature Application.” *Energy Conversion and Management: X*, vol. 24, article 100722 (2024). [DOI]

SELECTED PROJECTS

Flash Joule Heating System for Turbostratic-Graphene Synthesis Mar 2023 – Sep 2023
Research Project / Supervisor: Dr. Md. Abdullah Al-Bari

- Designed and built a flash Joule heating system using a 70 mF capacitor bank and a custom AC–DC converter for charging to 300 V.
- Implemented Arduino-based switching for approximately 30 ms discharge pulses to rapidly heat carbon-black feedstock upto 3000K temperature.

KUET Mars Rover (Team Durbar) Mar 2022 – Feb 2023
Sub-Team Lead, Control Systems and Software

- Led control/software development for a rocker-bogie rover; developed its drive system and wrote C++ code for control.
- Designed and constructed a manipulator arm capable of lifting over 5 kg.

Graduate course projects 2025 – 2026

- Robot Motion Planning using EB-RRT**: Implemented time-based RRT with offline elastic-band optimization in Python, reducing path cost and improving smoothness across three simulated maps with static and moving obstacles.
- Wi-Fi-Based Human Counting**: Developed a Transformer with a convolutional stem to classify human counts from WiMANS CSI data, achieving 41.36% peak test accuracy in the unseen Empty Room environment.

TECHNICAL SKILLS

Robotics: Robot Kinematics/Dynamics, Impedance and Force control, Reinforcement Learning, Motion Planning, Robot simulation (MuJoCo), Franka Research 3, Universal Robots, Arduino, Raspberry Pi, STM32, Pixhawk/ArduPilot, ROS2, TensorFlow, OpenCV, Python, C/C++, MATLAB

Tools/Softwares: SolidWorks, AutoCAD, Blender, LaTeX, Inkscape, kdenlive, Adobe Illustrator

STANDARDIZED TEST SCORES

- IELTS (Dec 27, 2024):** Overall 7 (Listening: 7.5, Reading: 8, Writing: 6, Speaking: 6)
- GRE (Oct 8, 2024):** Total 302 (Quantitative: 165, Verbal: 137, AWA: 3.5)

TEACHING AND TECHNICAL MENTORING

Summer Intern Mentoring Summer 2026
Medical Robotic Systems Lab, METU

- Mentored two summer interns on motorizing a laparoscopic gripper module for planned integration with a UR5e robot arm for robot-assisted laparoscopic manipulation.

FabLab KUET May 2022 – Apr 2023
Student Operator

- Trained more than 100 students in fabrication technologies, supported project prototyping, and operated 3D printers, CNC machines, and laser cutters.

LOOP Apr 2020 – Apr 2022
Project Manager

- Organized Arduino-based robotics workshops for over 100 students and mentored student teams in robot design and programming.

CADers Jun 2019 – Mar 2023
Information and Technology Officer

- Organized AutoCAD and SolidWorks workshops for over 200 students and developed CAD tutorials to support student learning.